

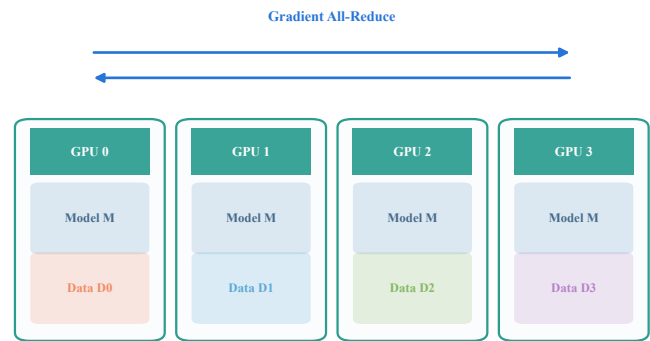
Distributed Training Mechanisms: What Is Sharded and How It Communicates

Mechanisms from Table 4-21 plus TP, PP, Ulysses Parallelism and Context Parallelism

Data Parallel (DDP)

ADOPT

replicate model; split mini-batch



Shard	batch only; complete model per GPU
Comm	bucketed gradient All-Reduce
U-MoE	adopt: bucket view + static graph

Overlap Grad Reduce

IN DDP

launch each bucket when gradients are ready

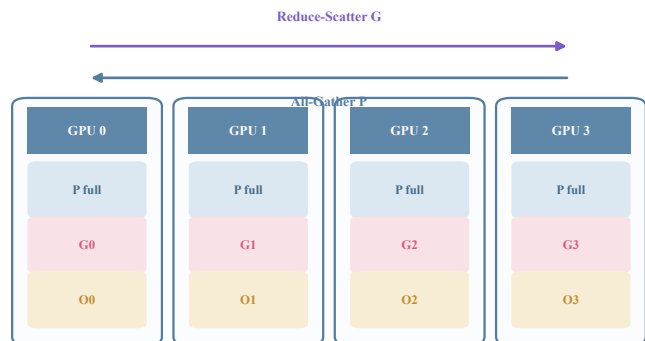


Shard	same DDP model and batch layout
Comm	async bucket All-Reduce vs backward
U-MoE	DDP reducer already provides it

Distributed Optimizer

NOT NOW

Megatron optimizer-state sharding

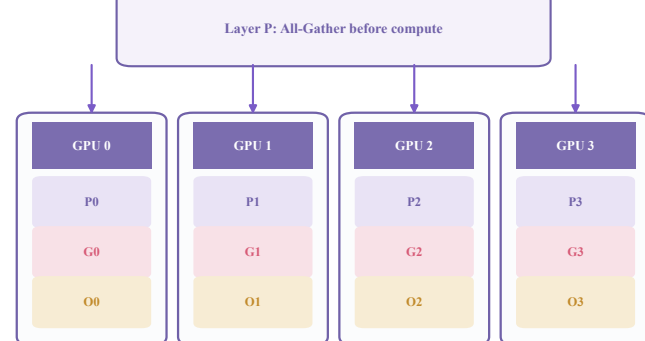


Shard	optimizer and gradient shards
Comm	Reduce-Scatter + parameter All-Gather
U-MoE	4.11 M params: little memory saved

ZeRO / FSDP

NOT NOW

shard P, G and O by stage

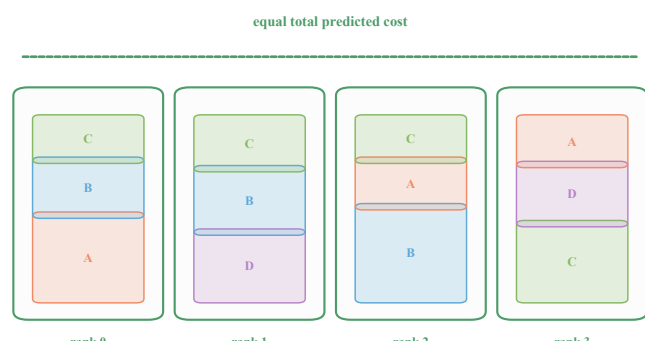


Shard	parameters / gradients / optimizer
Comm	per-layer All-Gather + Reduce-Scatter
U-MoE	activation peak dominates; extra collectives

Cost-Aware Data Shard

PRINCIPLE

equalize predicted work on every rank

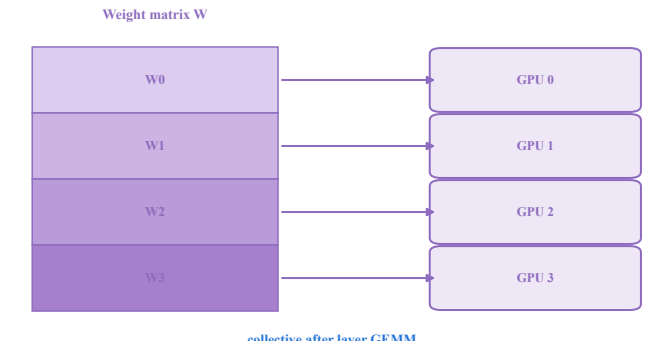


Shard	samples; fixed model on every rank
Comm	ordinary DDP after balanced assignment
U-MoE	use only with measured cost heterogeneity

Tensor Parallel (TP)

NOT NOW

split matrix dimensions within each layer

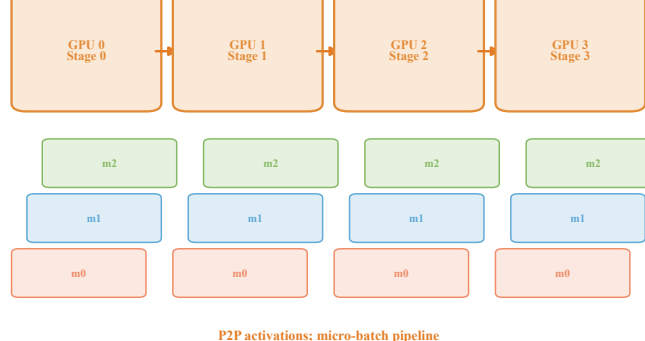


Shard	rows / columns of every large tensor
Comm	All-Reduce or All-Gather each layer
U-MoE	C=96 matrices too small for TP

Pipeline Parallel (PP)

NOT NOW

place consecutive layer stages on GPUs

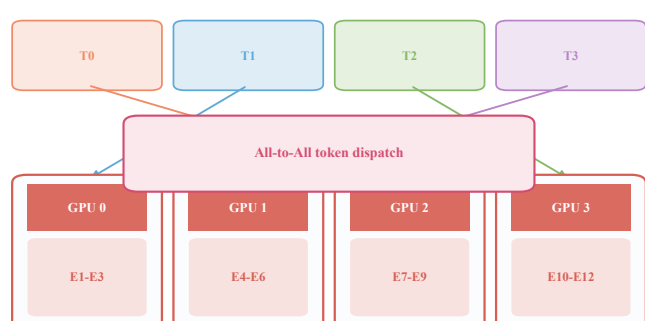


Shard	contiguous layer stages
Comm	P2P activations and gradients
U-MoE	shallow backbone; bubble dominates

Expert Parallel (EP)

NOT NOW

place different experts on different GPUs

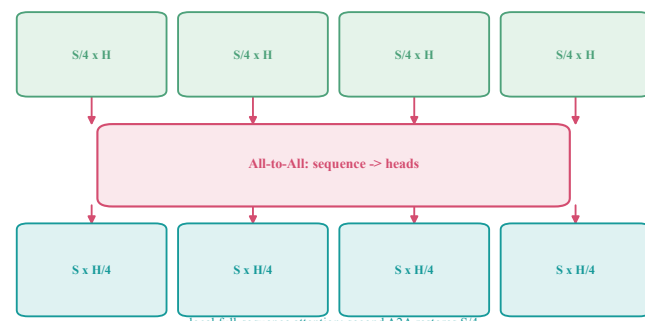


Shard	expert weights and routed tokens
Comm	All-to-All dispatch + combine
U-MoE	12 small experts already fit locally

Ulysses Parallelism (UP)

NOT NOW

swap sequence shards for head shards

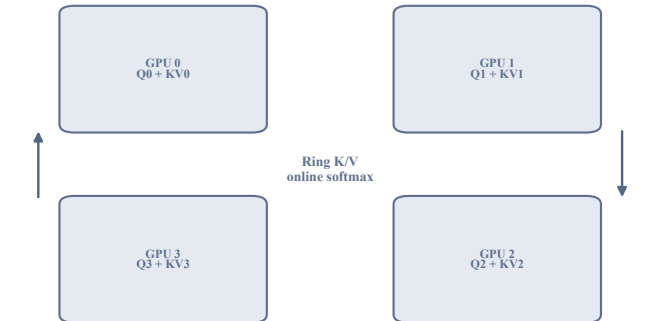


Shard	sequence first, then attention heads
Comm	two All-to-All operations per layer
U-MoE	window sequence is only 64 tokens

Context Parallel (CP)

NOT NOW

keep local Q; rotate K/V context blocks



Shard	sequence context and K/V blocks
Comm	ring P2P K/V exchange
U-MoE	local 8 x 8 windows need no CP

Communication Legend



Measured Profile

4.11 M parameters; < 0.1 GB model states
61-76 GB peak is activation / loss / MoE capacity

Recommended for U-MoE-Fusion

DDP + bucket view + static graph + 8 MiB buckets + fused Adam
Cost-aware data assignment only after measured heterogeneity

Why Other Axes Wait

No parameter, expert-capacity, depth, or long-context bottleneck
Extra collectives would target the wrong resource